

Univariate Optimization

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Outline

Optimization Problems

Local optimization

Fastest local optimization

A Fleeting Glimpse to Global Optimization

Wrap up & References

Solutions

- ▶ X any set, $f : X \rightarrow \mathbb{R}$ any function: optimization problem

$$(P) \quad f_* = \min \{ f(x) : x \in X \}$$

- ▶ Impossible (X inaccessible cardinal, f non computable function, ...)

- ▶ Let's start "easy":

- ▶ X "very easy": $X = \mathbb{R}$ or (even better) bounded $X = [x_-, x_+] \subset \mathbb{R}$

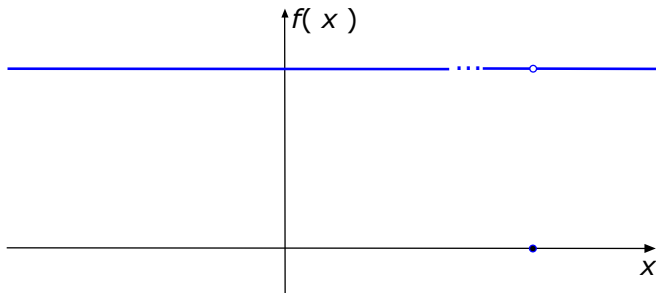
- ▶ an (efficient, pointwise) oracle for f available:

$\forall x \in X$, $f(x)$ is "easy to compute" (say, $O(1)$)

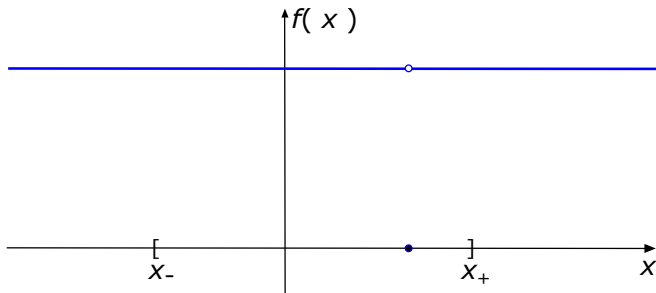
- ▶ Still not easy at all, in fact impossible in general [3, p. 408]

- ▶ Too trivial for $f(\cdot)$ linear or quadratic, $O(1)$ formulæ

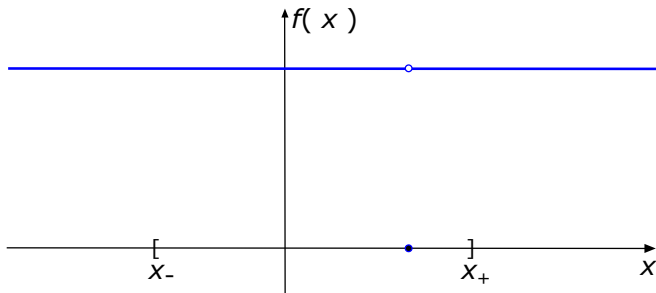
- ▶ Need to find a middle ground (one must $\exists$)



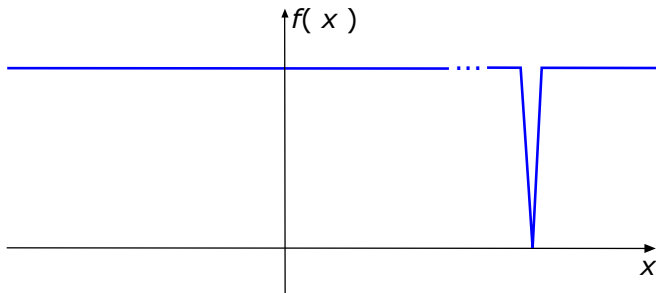
- Impossible because isolated minima can be anywhere [3, p. 408]



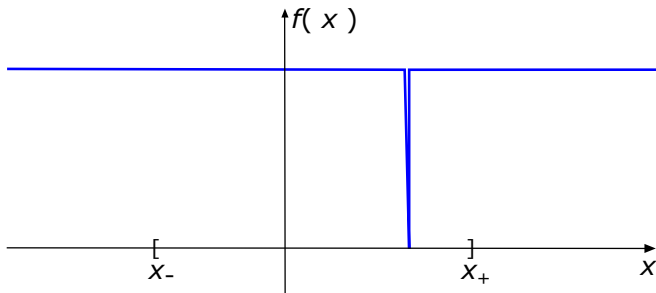
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- No: still uncountably many points to try



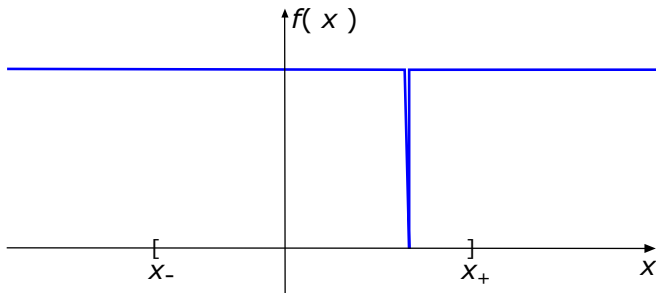
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- ▶ Is it because f “jumps”? No, f can have isolated $\downarrow$ spikes anywhere ... even on $X = [x_-, x_+]$ as spikes can be arbitrarily narrow
- ▶ Making it possible $\equiv$ impose speed limits on the rate of change

- ▶ **Impose spikes can't be arbitrarily narrow** $\equiv f$ **cannot change too fast** $\equiv f$ **Lipschitz continuous (L-c) on X** [4, p. 624]: $\exists L > 0$ s.t.

$$|f(x) - f(z)| \leq L|x - z| \quad \forall x, z \in X$$
- ▶ f **globally L-c** $\equiv X = \mathbb{R}$, **locally L-c at x** $\equiv \exists \varepsilon > 0$ s.t. $X = [x - \varepsilon, x + \varepsilon]$
- ▶ Note: L **depends on X** (locally L-c $\not\Rightarrow$ globally L-c)
- ▶ $f : \mathbb{R} \rightarrow \mathbb{R}$ **continuous at x** $\equiv \forall \{x_i\} \rightarrow x \implies \{f(x_i)\} \rightarrow f(x) \equiv \forall \varepsilon > 0 \exists \delta > 0$ s.t. $z \in [x - \delta, x + \delta] \implies |f(z) - f(x)| \leq \varepsilon$
- ▶ continuous on $X \equiv \forall x \in X$, just “continuous” $\equiv X = \mathbb{R} \equiv f \in C^0$
- ▶ Many “simple” functions C^0 + continuity easily preserved:
 $f, g \in C^0 \implies f + g, f \cdot g, \max\{f, g\}, \min\{f, g\}, f(g(\cdot)) \in C^0$
- ▶ f locally L-c at $x \implies f$ continuous at x (**check**)

Exercise: Come up with f locally L-c everywhere but not globally L-c

Exercise: Come up with f continuous but not L-c on some finite $X = [x_-, x_+]$

- ▶ Still need to impose $X = [x_-, x_+]$ with $D = x_+ - x_- < \infty$ (finite diameter), otherwise isolated $\downarrow$ spikes need not even be “very narrow”
- ▶ f L -c $\implies$ one ε -optimum can be found with $O(LD / \varepsilon)$ evaluations:
uniformly sample X with step $2\varepsilon / L$ [3, p. 411]

Exercise: Prove the above

- ▶ **Bad news:** no algorithm can work in less than $\Omega(LD / \varepsilon)$ [3, p. 413]
(proof uses **adversarial function**, **not typical** in learning applications)
- ▶ # steps inversely proportional to accuracy, just not doable for “small” ε
- ▶ Even very dramatically **worse** if $X \subset \mathbb{R}^n$ (will see)
- ▶ **No free lunch theorem says** “all algorithms equally bad” [7], i.e.,
“if an algorithm is very good in some cases it has to be very bad in others”
- ▶ Also, L **generally unknown** and **not easy to estimate** (will see)
but **algorithms actually require/use it**

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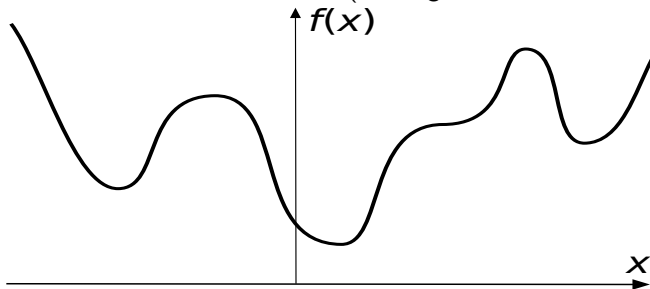
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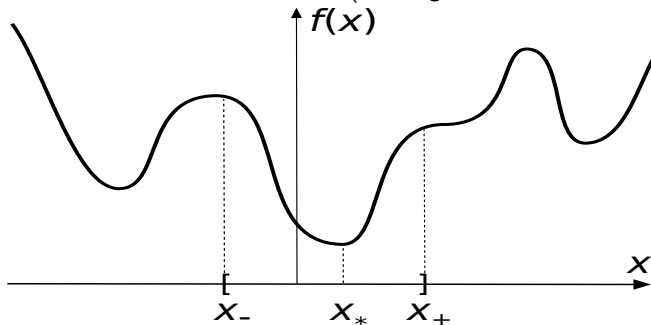
Solutions

- ▶ Even if I should stumble in x_* , how do I recognize it?
- ▶ Turns out this is “the really difficult thing” (cf. knowing f_*)
- ▶ Simpler to start with a weaker condition: x_* is local minimum if
$$x_* = \operatorname{argmin} \{ f(x) : x \in X(x_*, \varepsilon) = [x_* - \varepsilon, x_* + \varepsilon] \} \quad \text{for some } \varepsilon > 0$$
- ▶ Stronger notion: strict local minimum if $f(x_*) < f(z) \quad \forall z \in X(x_*, \varepsilon) \setminus \{x_*\}$
- ▶ Why useful? Because “near x_* , f typically has a predictable shape”
- ▶ f (strictly) unimodal on $X = [x_-, x_+]$:
 - ▶ has minimum $x_* \in X$
 - ▶ is (strictly) decreasing in $[x_-, x_*]$ and increasing in $[x_*, x_+]$
- ▶ x_* local minimum $\implies$ typically $\exists \varepsilon > 0$ s.t. f (strictly) unimodal on $X(x_*, \varepsilon)$

- Most functions are **not unimodal** (although **some are**, will see)

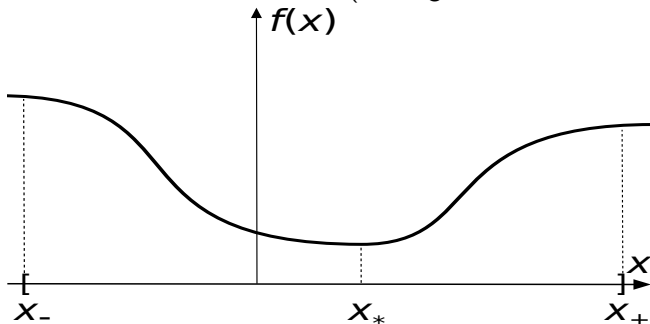


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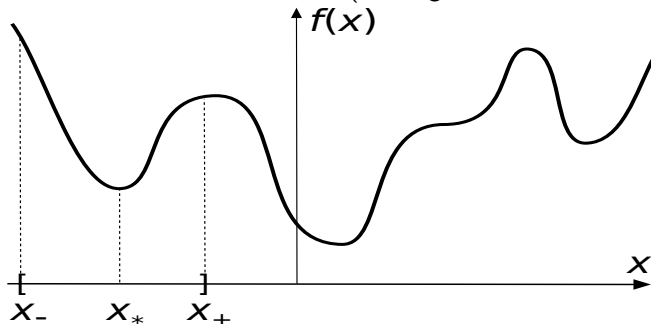
- ▶ But they **are** if you focus on the **attraction basin of x_*** and

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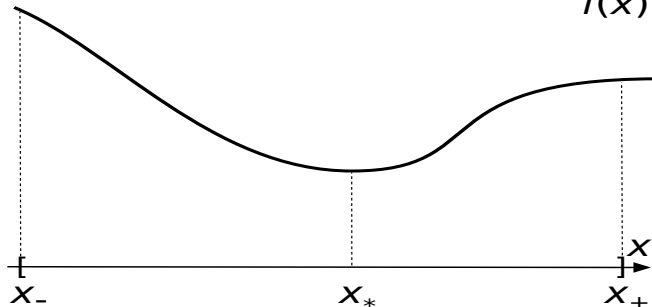
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- ▶ Unfortunately, this is true **for every local optimum**

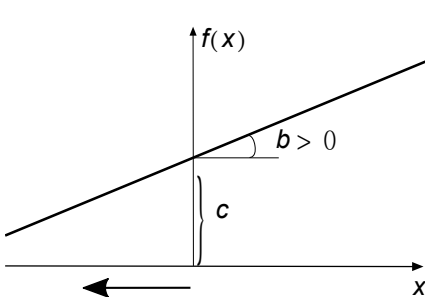
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- ▶ But they **are** if you focus on the **attraction basin of x_*** and **restrict there**
- ▶ Unfortunately, this is true **for every local optimum**
- ▶ All local optima “look the same”, **comprised the global one**
- ▶ Yet, this makes it finding **some** local optimum **a lot easier**
- ▶ **Finding the right (global) one another matter entirely**

- ▶ Where is the local minimum? In the direction where f is decreasing
- ▶ If we could see this directly we could get there fast

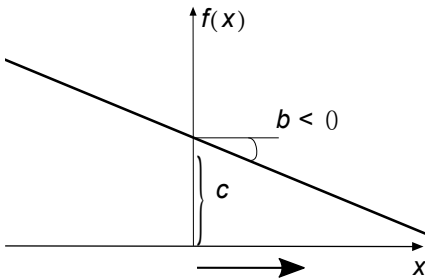
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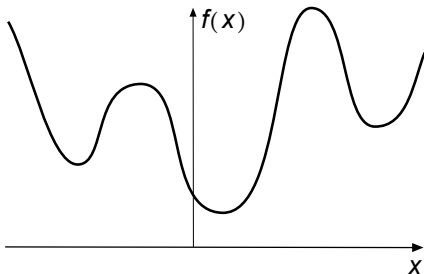
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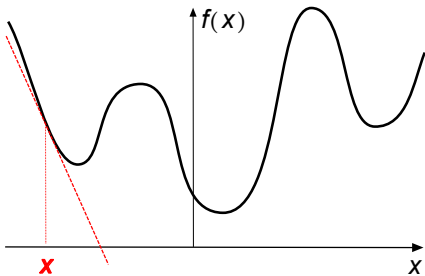


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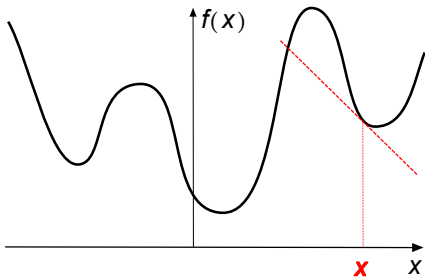
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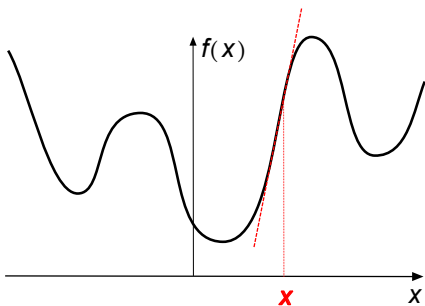
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always left if $b > 0$, right if $b < 0$
- ▶ f nonlinear $\implies$ first-order model of f
at x : $L_x(z) = f'(x)(z - x) + f(x)$
- ▶ best linear approximation of f at x :
 $L_x(z) \approx f(z) \quad \forall z \in [x - \epsilon, x + \epsilon]$
for some (small) $\epsilon > 0$

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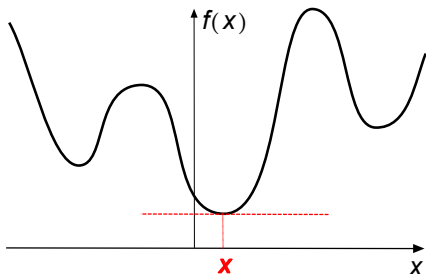
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for some (small) $\epsilon > 0$
- ▶ Trusty old (first) derivative $f'(x)$ [6, §2.3]
- ▶ $f'(x) =$ slope of the tangent line to the graph of f in x :
 $f'(x) < 0 \implies f$ decreasing at x ,

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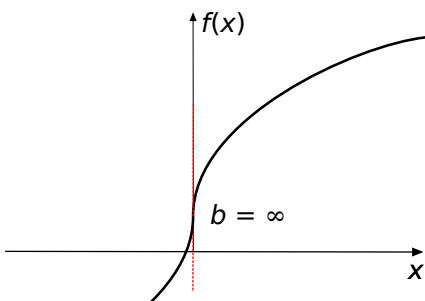
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 $f'(x) < 0 \implies f$ decreasing at x , $f'(x) > 0 \implies f$ increasing at x
- ▶ x_* local minimum $\simeq f'(x_*) = 0 \equiv$ root of $f' \equiv$ stationary point

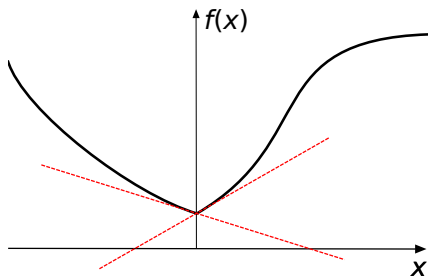
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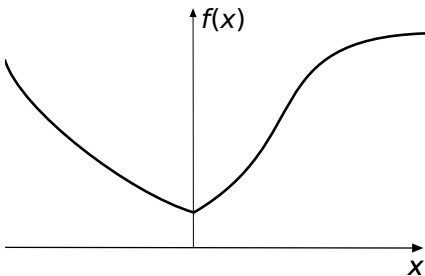
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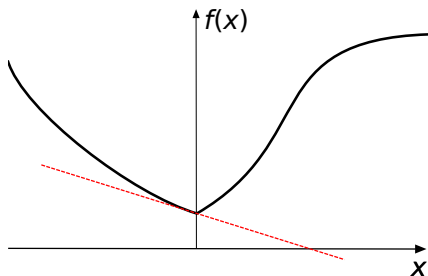
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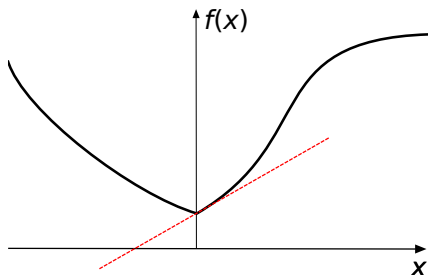


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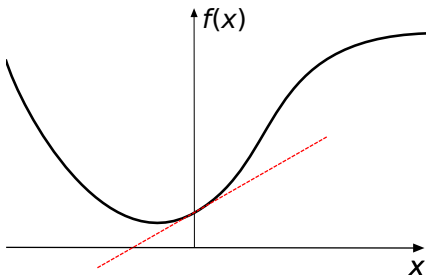
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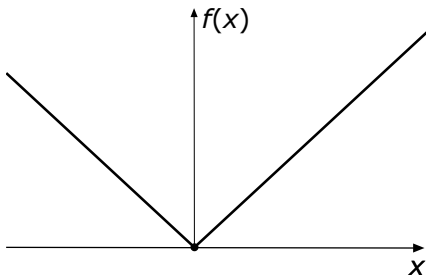
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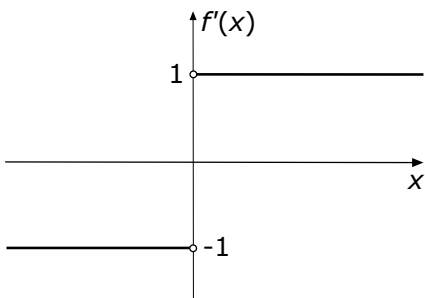
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- ▶ f **differentiable** at x if $f'(x) \ni \text{finite} \equiv f'_-(x) = f'_+(x) (\Leftarrow \ni \text{finite})$
- ▶ **Nondifferentiable functions happen in practice:** $f(x) = |x| = \max\{x, -x\}$

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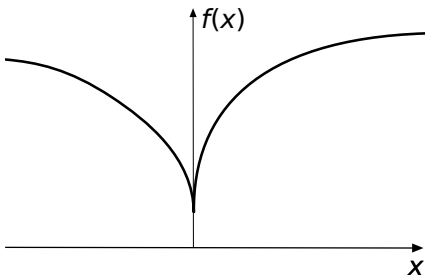
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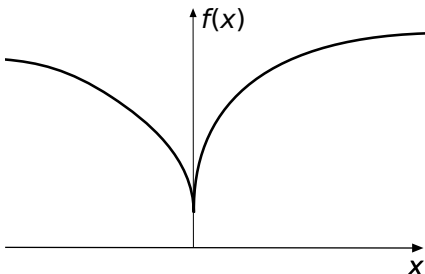
► $f'(x) = -1$ if $x < 0$, $f'(x) = +1$ if $x > 0$, $f'(x) = ???$ if $x = 0$

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- ▶ f differentiable at x if $f'(x) \exists$ finite $\equiv f'_-(x) = f'_+(x)$ ($\Leftarrow \exists$ finite)
- ▶ Can be as different as $-\infty$ and $+\infty$

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- ▶ f **differentiable** at x if $f'(x) \exists$ **finite** $\equiv f'_-(x) = f'_+(x)$ ($\Leftarrow \exists$ finite)
- ▶ Nondifferentiable $\equiv f'$ **not continuous**
- ▶ f **differentiable** at $x \implies f$ **continuous** at x , but $\Leftarrow$ **does not hold**

Exercise: Prove it

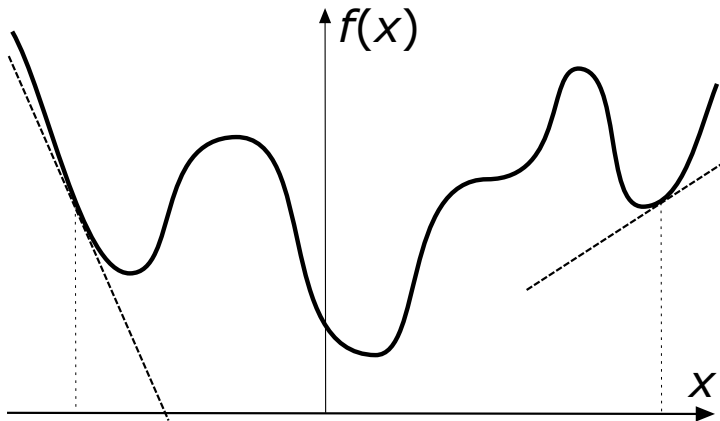
- ▶ $f' \in C^0 \equiv f \in C^1 \equiv f$ continuously differentiable $\implies f \in C^0$
- ▶ $f'' \in C^0 \equiv f \in C^2 \equiv f' \in C^1 \implies f' \in C^0 \implies f \in C^1 \implies f \in C^0$
- ▶ $f \in C^1$ globally L-c on (open) $X \implies |f'(x)| \leq L \quad \forall x \in X$

Exercise: Prove it, is $\Leftarrow$ true?

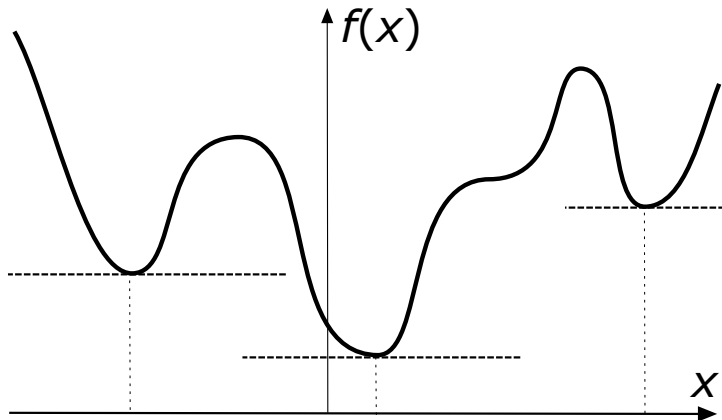
Exercise: Formally prove $\exists f \in C^0$ but not L-c on some finite $X = [x_-, x_+]$

- ▶ Extreme value theorem [6, Th. 2.2.9]: $f \in C^0$ on $X = [x_-, x_+]$ (closed) finite $\implies \max \{ f(x) : x \in X \} < \infty$, $\min \{ f(x) : x \in X \} > -\infty$
- ▶ $f \in C^1$ on X finite (closed) $\implies f$ globally L-c on X
- ▶ Best possible case ever: $f \in C^2$ (actually, C^3) on finite $X \implies$ both f and f' globally L-c on X

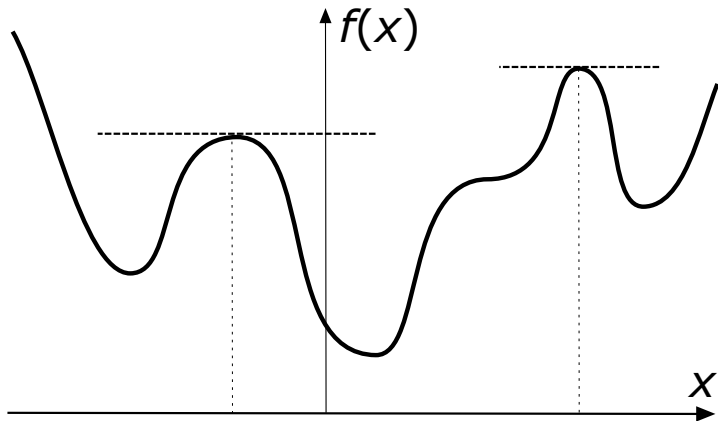
- ▶ Derivatives of many simple functions are known, (**almost** always) continuous
 - ▶ $[x^k]' = kx^{k-1}$
 - ▶ $[e^x]' = e^x$, $[\ln(x)]' = 1/x$
 - ▶ $[\sin(x)]' = \cos(x)$, $[\cos(x)]' = -\sin(x)$
- ▶ Many functional operations (**almost** always) preserve differentiability
 - ▶ $[\alpha f(x) + \beta g(x)]' = \alpha f'(x) + \beta g'(x)$
 - ▶ $[f(x) \cdot g(x)]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
 - ▶ $[f(x)/g(x)]' = [f'(x) \cdot g(x) - f(x) \cdot g'(x)] / g(x)^2$
 - ▶ $[f(g(x))]' = f'(g(x)) \cdot g'(x)$ (**chain rule**)
- ▶ A few common functional operations **don't**:
$$\max\{f(x), g(x)\} \quad , \quad \min\{f(x), g(x)\}$$
- ▶ In general **automatic differentiation well-developed, available, fast** [8]
 $\implies$ **actually** (writing code to) **compute derivatives not our business**



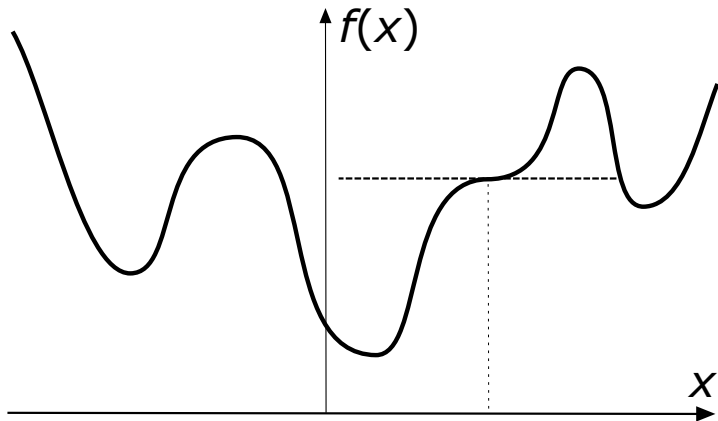
- If $f'(x) < 0$ or $f'(x) > 0$, x clearly cannot be a local minimum



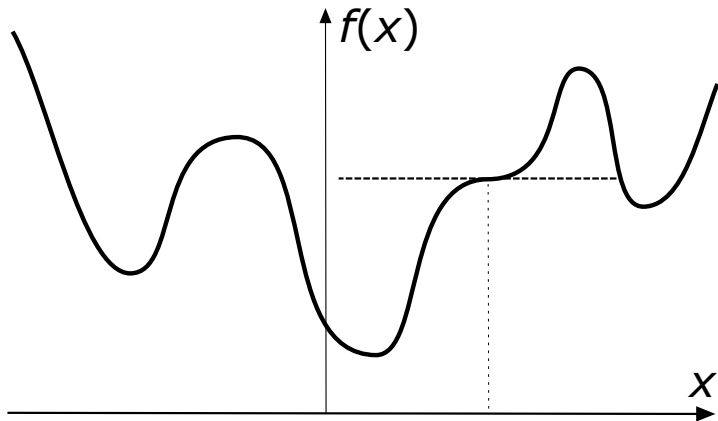
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- ▶ However, $f'(x) = 0$ also in local (hence global) maxima



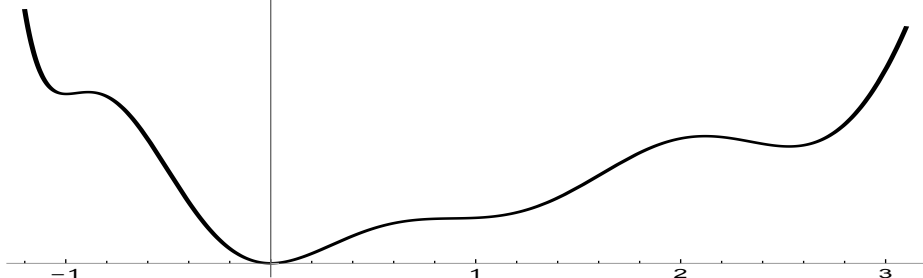
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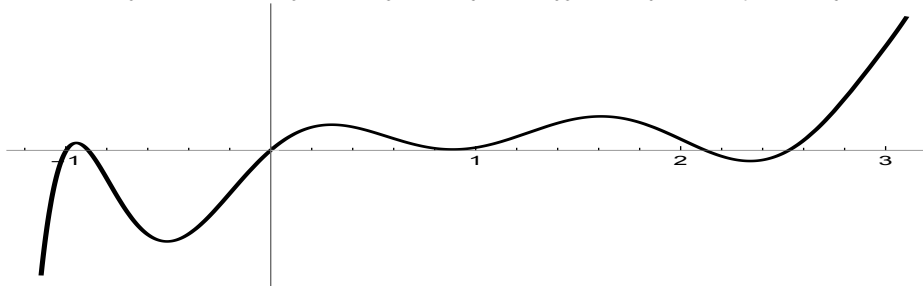
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... as well as in saddle points
- ▶ How do I **tell them apart**? Look at $f'' = [f']' =$ second derivative

A polynomial example: roots of f' are the “interesting” points 12

$$f(x) = \frac{91}{30}x^2 - \frac{19}{6}x^3 - \frac{54}{25}x^4 + \frac{93}{23}x^5 - \frac{23}{36}x^6 - \frac{121}{93}x^7 + \frac{72}{91}x^8 - \frac{13}{74}x^9 + \frac{9}{640}x^{10}$$



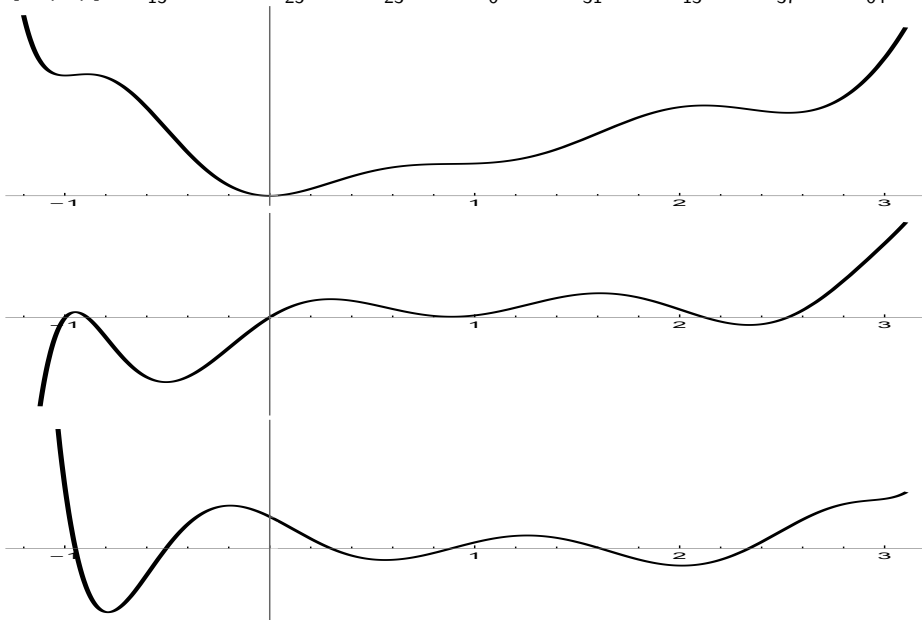
$$f'(x) = \frac{91}{15}x - \frac{19}{2}x^2 - \frac{216}{25}x^3 + \frac{465}{23}x^4 - \frac{23}{6}x^5 - \frac{847}{93}x^6 + \frac{576}{91}x^7 - \frac{117}{74}x^8 + \frac{9}{64}x^9$$



The sign of f'' (if not 0) tells apart maxima from minima

13

$$[f'(x)]' = \frac{91}{15} - 19x - \frac{648}{25}x^2 + \frac{1860}{23}x^3 - \frac{115}{6}x^4 - \frac{1694}{31}x^5 + \frac{576}{13}x^6 - \frac{468}{37}x^7 - \frac{81}{64}x^8$$



- ▶ In **simple cases**, you get the answer by a **closed formula** (surprised?)
- ▶ $f(x) = bx [+c]$ (linear), $f'(x) = b = 0 \implies \nexists x$ if $b \neq 0$, $\forall x$ if $b = 0$
- ▶ $f(x) = ax^2 + bx [+c]$ (quadratic, $a \neq 0$), $f'(x) = 2ax + b = 0 \implies x = -b/2a$ **unique minimum if $a > 0$, maximum if $a < 0$**
- ▶ Generalise almost only to polynomials whose root have a closed formula (degree 3, some degree 4)
- ▶ Little hope for most trascendental / trigonometric / mixed unless you are **very lucky**
- ▶ Need an **algorithm** for solving **nonlinear** equations

- ▶ f' continuous + intermediate value theorem [6, Th. 2.2.10] $\implies$
 $f'(x_-) < 0 \wedge f'(x_+) > 0 \implies \exists x \in [x_-, x_+] \text{ s.t. } f'(x) = 0$
- ▶ Theorems breed algorithms: dichotomic search

```

procedure  $x = DS(f, x_-, x_+, \varepsilon)$ 
  do forever // invariant:  $f'(x_-) < -\varepsilon, f'(x_+) > \varepsilon$ 
     $x \leftarrow \text{in\_middle\_of}(x_-, x_+)$ ; compute  $f'(x)$ ;
    if ( $|f'(x)| \leq \varepsilon$ ) then break;
    if ( $f'(x) < 0$ ) then  $x_- \leftarrow x$ ;
    else  $x_+ \leftarrow x$ ;

```

- ▶ Trivial choice: $\text{in_middle_of}(x_-, x_+) \{ \text{return}((x_+ + x_-) / 2) \}$
- ▶ Linear convergence with $r = 0.5 < 0.618 \implies$
 $k \approx 3.32 \log(D / \delta) < 4.78 \log(D / \delta)$ (err, who is δ ?)
- ▶ f' L-c with constant $L \equiv \text{L-smooth} \implies k \approx 3.32 \log(LD / 2\varepsilon)$ (check)
- ▶ Does it show in practice?

- ▶ What if the assumption is not satisfied?

- ▶ Obvious solution:

$\Delta x \leftarrow 1;$	// or whatever value > 0
while $(f'(x_+) \leq -\varepsilon)$ do	
$x_+ \leftarrow x_+ + \Delta x; \Delta x \leftarrow 2\Delta x;$	// or whatever factor > 1

- ▶ Of course, the same “in reverse” for x_- ($\Delta x = -1$)
- ▶ Will work in practice for all “reasonable” function
- ▶ Works if f **coercive**: $\lim_{|x| \rightarrow \infty} f(x) = \infty$

Exercise: construct an example where x_+ / x_- exist but are not found

- ▶ If $f_* = -\infty$, $x_{\pm}$ may $\rightarrow \pm\infty$ “proving” unboundedness ($f(x_{\pm}) \rightarrow -\infty$)
but how do you stop? (need a “finite $-\infty$ ”)

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Solutions

- ▶ Choosing x “right in the middle” just the simplest approach:
better if x is close to x_* (ideally, $x = x_*$ would stop in one iteration)
- ▶ One knows a lot about f : $f(x_-)$, $f(x_+)$, $f'(x_+)$, $f'(x_-)$, let's use that
- ▶ Powerful general idea: construct a model of f based on known information
- ▶ Quadratic interpolation: $ax^2 + bx + c$ that “agrees” with f at x_+ , x_-
- ▶ Three parameters, four conditions, something's gotta give (three cases)
- ▶ One way: $2ax_+ + b = f'(x_+)$, $2ax_- + b = f'(x_-) \implies$

$$a = \frac{f'(x_+) - f'(x_-)}{2(x_+ - x_-)}, \quad b = \frac{x_+ f'(x_-) - x_- f'(x_+)}{x_+ - x_-}$$

- ▶ Minimum solves $2ax + b = 0$ (c irrelevant) $\equiv$

$$x = \frac{x_- f'(x_+) - x_+ f'(x_-)}{f'(x_+) - f'(x_-)} \quad \begin{array}{l} \text{“method of false position”} \\ \text{a.k.a. “secant formula”} \end{array}$$

always in the middle between x_+ and x_- (check)

Exercise: develop the other cases of quadratic interpolation and discuss them

- ▶ Very general issue: **the model is an estimate $\implies$ wrong $\implies$ bad choices**
- ▶ In this case, the model can be “very skewed”:
$$f'(x_+) \gg -f'(x_-) \implies x \approx x_- \quad , \quad f'(x_+) \ll -f'(x_-) \implies x \approx x_+$$
- ▶ Can lead to **very short steps $\implies$ slow convergence**
- ▶ General remedy: **never completely trust the model** $\equiv$ regularise, stabilise, ...
- ▶ In this case: **minimum guaranteed decrease $\sigma \leq 0.5$** (safeguard)
$$x \leftarrow \max\{x_- + \sigma(x_+ - x_-), \min\{x_+ - \sigma(x_+ - x_-), x\}\}$$
- ▶ Worst case: linear convergence with $r = 1 - \sigma$
- ▶ Hopefully (much) **faster than that when the model is “right”**
- ▶ Does it really show in practice? And how much faster?

- ▶ Quadratic interpolation has superlinear convergence if started “close enough”:
[5, Th. 2.4.1] $f \in C^3$, $f'(x_*) = 0$ and $f''(x_*) \neq 0 \implies$
 $\exists \delta > 0$ s.t. $x^0 \in [x_* - \delta, x_* + \delta] \implies \{x^i\} \rightarrow x_*$ with $p = (1 + \sqrt{5}) / 2$
($1 < p = g \approx 1.618 < 2$, don't you just love maths?)
- ▶ This proves “very fast” already, but can we make it even faster?

- ▶ Quadratic interpolation has **superlinear** convergence **if started “close enough”**:
[5, Th. 2.4.1] $f \in C^3$, $f'(x_*) = 0$ and $f''(x_*) \neq 0 \implies$
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($1 < p = g \approx 1.618 < 2$, don't you just love maths?)
- ▶ This proves “very fast” already, but can we make it **even faster**?
- ▶ **Four** conditions $\implies$ can **fit a cubic polynomial** and use its minima
- ▶ Rather tedious to write down, analyse and implement [5, § 2.4.2][4, p. 57]
- ▶ Theoretically pays: cubic interpolation has **quadratic** convergence ($p = 2$)
- ▶ Seems to work pretty well in practice

Exercise: (not for the faint of heart): develop cubic interpolation

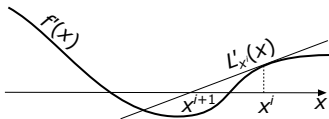
- ▶ Better model of $f \equiv f' \implies$ better guess of x_* $\implies$ faster
- ▶ Better model $\iff$ either more points or more (higher-order) derivatives
- ▶ Newton's method (tangent method): first-order model of f' at x^i

$$L'_i(x) = L'_{x^i}(x) = f'(x^i) + f''(x^i)(x - x^i) \approx f'(x)$$

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$$L'_i(x) = L'_{x^i}(x) = f'(x^i) + f''(x^i)(x - x^i) \approx f'(x)$$
- ▶ Solve $L'_i(x) = 0 \approx f'(x) = 0 \quad \equiv$

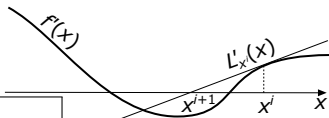
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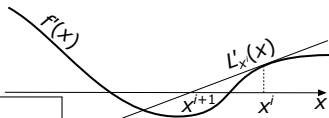
```

procedure  $x = NM(f, x, \varepsilon)$ 
  while  $|f'(x)| > \varepsilon$  do
     $x \leftarrow x - f'(x) / f''(x);$  // what if  $f''(x) = 0$ ?
  
```

- ▶ Better model of $f \equiv f' \implies$ better guess of x_* $\implies$ faster
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```

procedure  $x = NM(f, x, \varepsilon)$ 
  while ( $|f'(x)| > \varepsilon$ ) do
     $x \leftarrow x - f'(x) / f''(x);$  // what if  $f''(x) = 0$ ?
    
```

- ▶ Alternative view (**check**): minimize second-order model of f at x^i
 $Q_i(x) = Q_{x^i}(x) = f(x^i) + f'(x^i)(x - x^i) + f''(x^i)(x - x^i)^2 / 2$
 (but Newton's actually a method to solve nonlinear equations)
- ▶ Converges fast (at all!) only if started "close enough" to x_* [1, Th. 8.2.3]
- ▶ Would require globalization (possible), will see in $\neq$ context

- Second-order Taylor's formula: $\forall z \exists w \in [x, z]$ s.t.

$$f(z) - L_x(z) = f''(w)(z - x)^2/2 \quad [6, \text{Th. 2.5.4}]$$

“the error of L_x in z is $(z - x)^2 \times$ the value of f'' somewhere in the middle”

- Hypotheses: $f \in C^3$, $f'(x_*) = 0$ and $f''(x_*) \neq 0$
- Thesis: $\exists \delta > 0$ s.t. $x^0 \in [x_* - \delta, x_* + \delta] \implies \{x^k\} \rightarrow x_*$ with $p = 2$
- Proof: $x^{i+1} - x_* = x^i - x_* + (f'(x_*) - f'(x^i)) / f''(x^i)$

$$= [f'(x_*) - f'(x^i) - f''(x^i)(x_* - x^i)] / f''(x^i)$$

Taylor's formula for f' : $\exists w \in [x^i, x_*]$ s.t.

$$f'(x_*) - f'(x^i) + f''(x^i)(x_* - x^i) = f'''(w)(x_* - x^i)^2/2$$

$$\implies x^{i+1} - x_* = [f'''(w) / 2f''(x^i)](x^i - x_*)^2$$

$\exists \delta > 0$ s.t. $|f''(x)| \geq k_2 > 0$ and $|f'''(w)| \leq k_1 < \infty$ (check)

$$\forall x, w \in [x_* - \delta, x_* + \delta] \implies |x^{i+1} - x_*| \leq [k_1 / 2k_2](x^i - x_*)^2$$

$$k_1(x^i - x_*) / 2k_2 \leq 1 \implies |x^{i+1} - x_*| < |x^i - x_*| \implies$$

$\{x^i\} \rightarrow x_*$ and the convergence is quadratic

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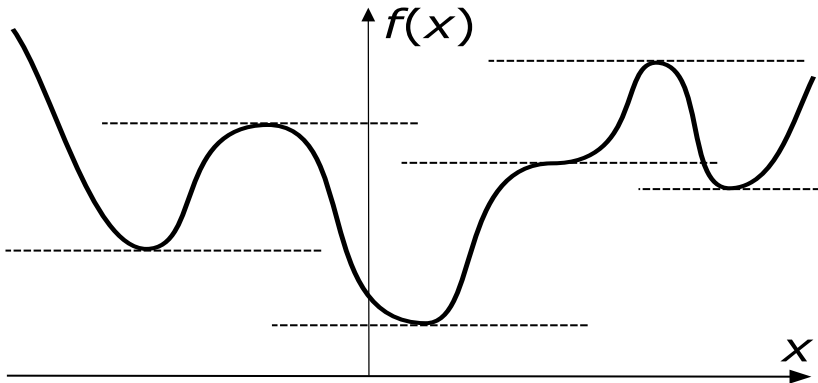
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- ▶ What does this all tells about **global** optimization?

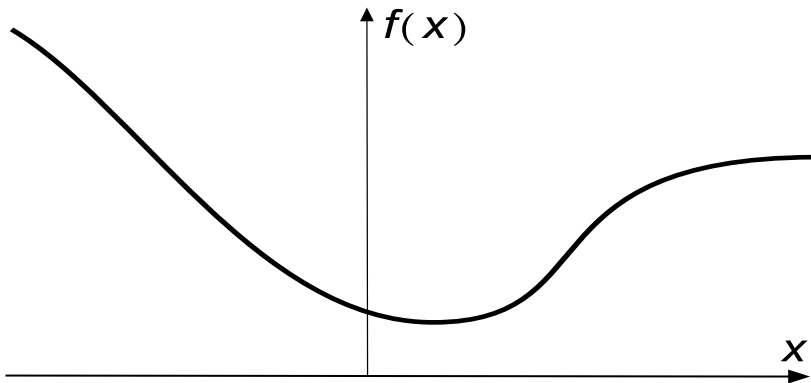
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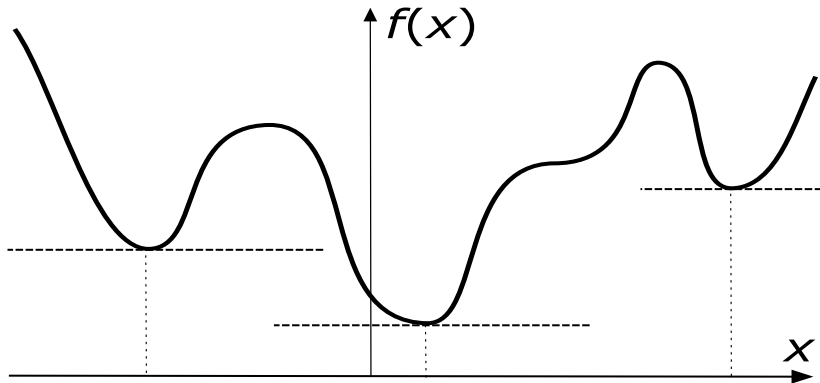
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- ▶ The obvious one would be unimodal, but **not easy to verify/construct**

- ▶ What does this all tell about **global** optimization?

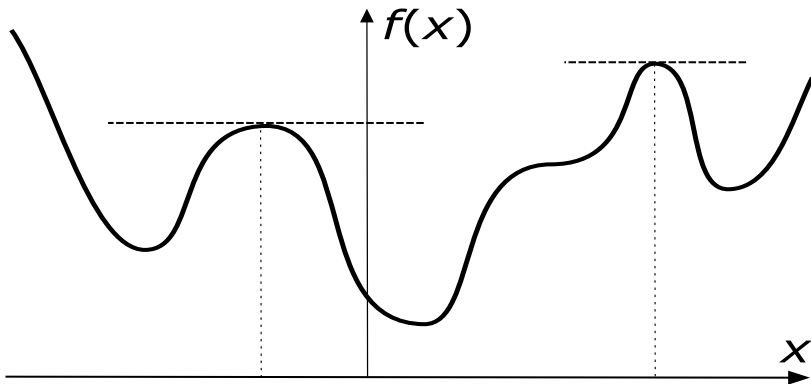
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- ▶ Intuitively: f has local **not global** minima

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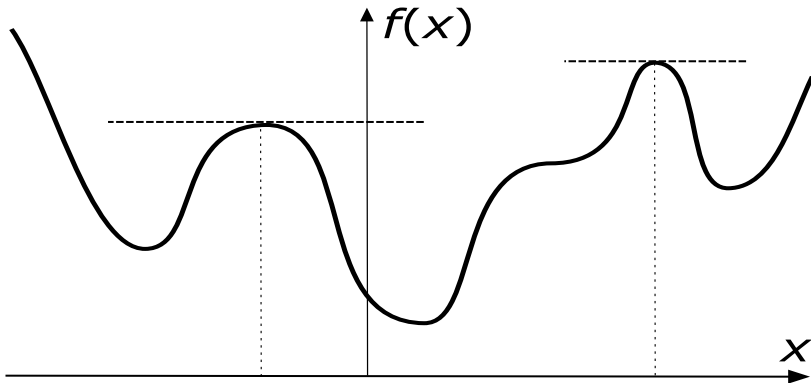
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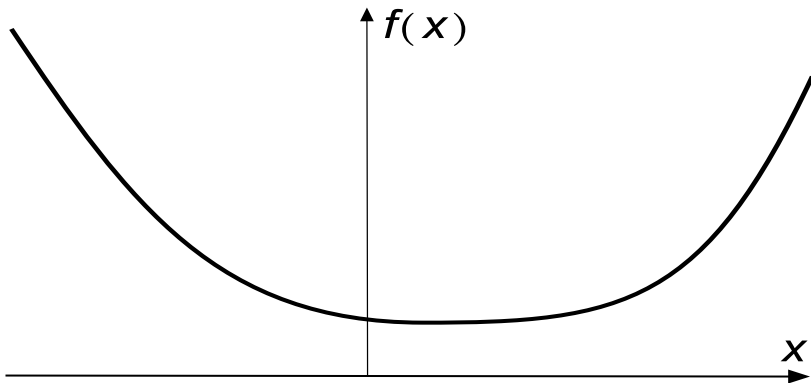
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- ▶ Intuitively: f has local **not global** minima $\implies$ has local **maxima**
- ▶ Avoid it: **stationary point** $\implies$ local minima $\equiv f'(x) = 0 \implies f''(x) \geq 0$

- ▶ What does this all tell about **global** optimization?

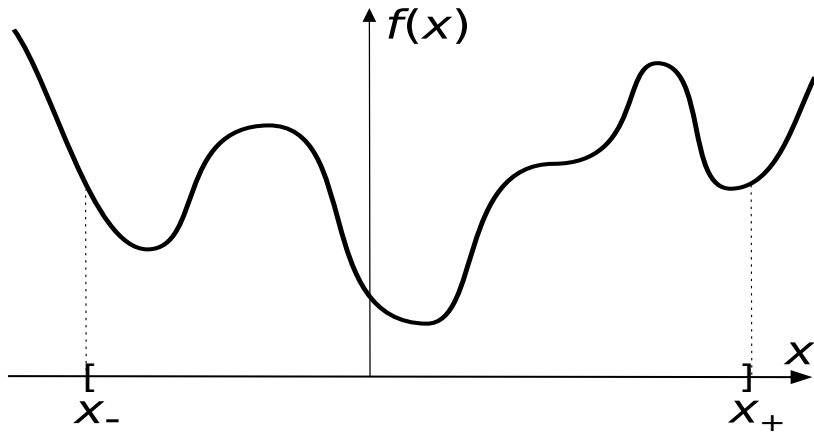
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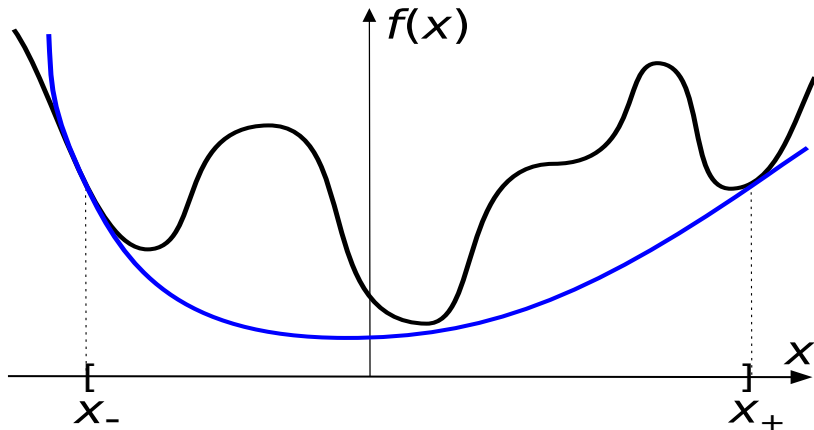
- ▶ Intuitively: f has local **not global** minima $\implies$ has local **maxima**
- ▶ Avoid it: **stationary point** $\implies$ local minima $\equiv f'(x) = 0 \implies f''(x) \geq 0$
- ▶ **Sufficient** condition: $f''(x) \geq 0 \forall x \in \mathbb{R} \implies f$ **convex**

- ▶ Convex $\simeq f'$ is monotone nondecreasing $\simeq f'' \geq 0$
- ▶ Not really because convex $\not\Rightarrow C^1$ (even less C^2), will see
- ▶ Some functions are convex + a few operations preserve convexity (will see)
 - $\Rightarrow$ the convex world is relatively large
 - $\Rightarrow$ can construct complicated (multivariate) convex functions/sets
- ▶ Plenty of theory [2] and software [10]
- ▶ Many models are purposely constructed convex (SVM) so that (global) optimization is easy
- ▶ “If you have the choice, choose convex”
- ▶ What if you don't and really need the global optimum?
- ▶ Will only say little here, but plenty of ways to satisfy your curiosity [9]

- Sift through all $X = [x_-, x_+]$, but using a clever guide

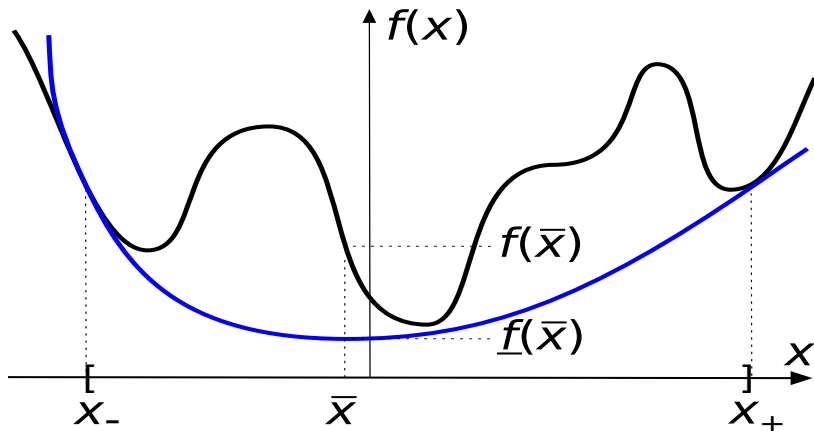


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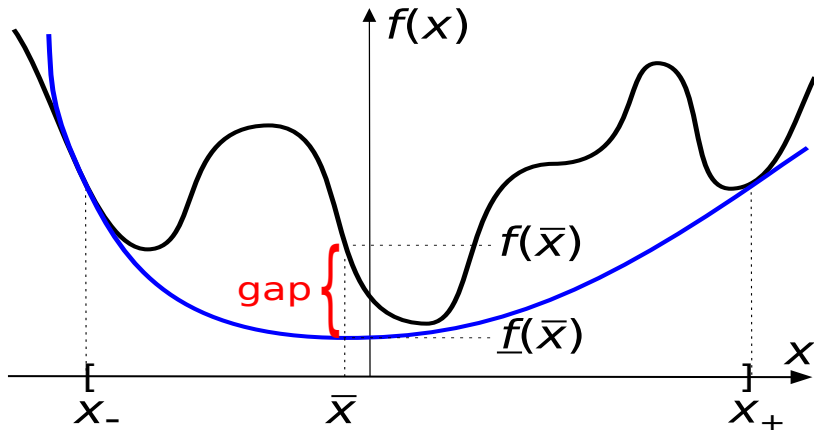
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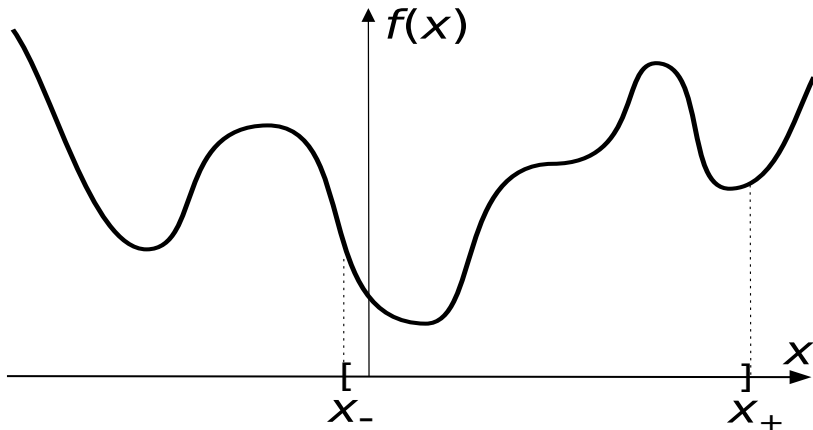
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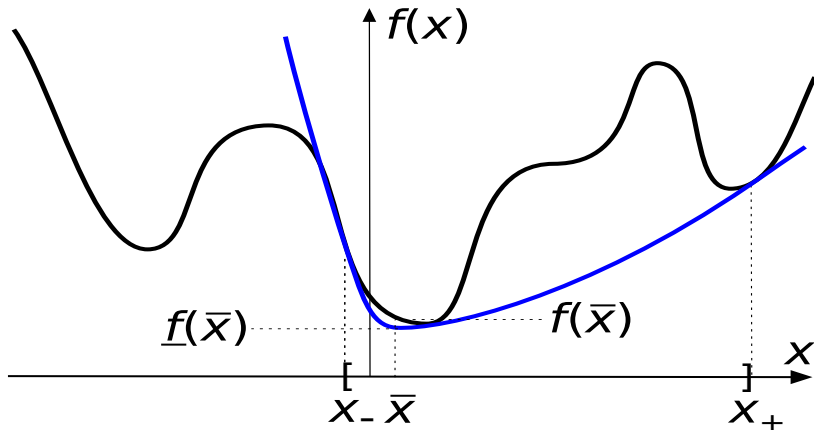
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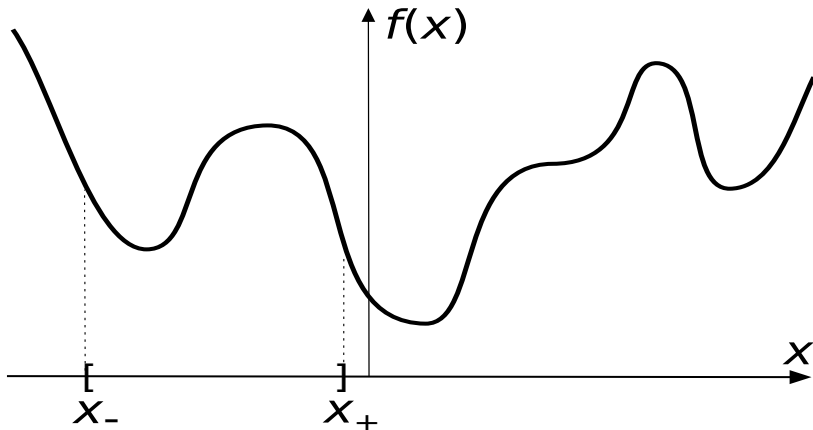
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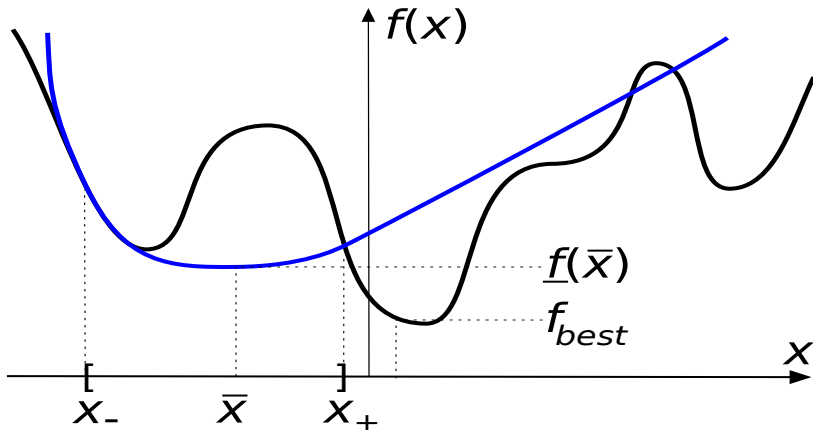
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- If gap $f(\bar{x}) - \underline{f}(\bar{x})$ too large, partition X and iterate
- $\underline{f}$ depends on partition, smaller partition (hopefully) $\implies$ better gap

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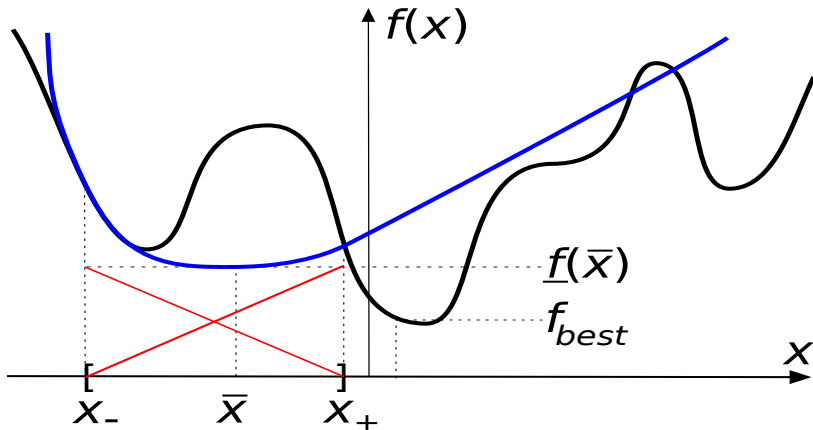
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- ▶ In a word? Surely **not in worst-case**:
keep dicing and slicing X until pieces “very small” $\implies$ exponential
- ▶ However, in practice it depends on:
 - ▶ “how much nonconvex” f really is
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- ▶ Mixed-Integer Linear Programs: all is “trivial” when integer fixed/relaxed

- ▶ In a word? Surely **not in worst-case**:
keep dicing and slicing X until pieces “very small” $\implies$ exponential
- ▶ However, in practice it depends on:
 - ▶ “how much nonconvex” f really is
 - ▶ how good $\underline{f}$ is as a lower approximation of f
- ▶ Clever approach: carefully choose your nonconvexities, e.g., integer variables
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- ▶ (Mixed-Integer) Nonlinear Nonconvex Programs: finding **any $\underline{f}$** complex
 - ▶ rewrite the expression of f in terms of unary/binary functions
 - ▶ apply specific convexification formulæ for each function
- ▶ Good news: implemented in available, well-engineered solvers and immensely less inefficient in practice than blind search
- ▶ Yet, immensely less efficient in practice than local optimization

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- ▶ Global (constrained or not) optimization **difficult** (impossible) in general
- ▶ **Local (unconstrained) optimization** much easier, useful in general:
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- ▶ Global (constrained or not) optimization **difficult** (impossible) in general
- ▶ **Local (unconstrained) optimization** much easier, useful in general:
once you know how to do local you can **try** global
- ▶ Algorithms are slow / medium / fast, “nicer” problems have faster algorithms
- ▶ The more **continuous** derivatives you have, the nicer the problem
- ▶ Derivatives $\implies$ first- and second-order **model**
- ▶ **f** “complicated”, model looks like **f** (close to **x**) and **simple**
- ▶ But **the map is not the world**, never **blindly** trust a model
- ▶ Fundamental concepts we will use all the time, let's move to **$n > 1$**

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- ▶ Take $\delta = \varepsilon / L$; then, $\forall y \in [x - \delta, x + \delta]$
 $|f(y) - f(x)| \leq L|x - y| \leq L\delta \leq L(\varepsilon / L) = \varepsilon$ [back]
- ▶ Note: we'll have a much simpler proof later, after we present the relationships between L-c and derivatives
 $f(x) = x^2$ is locally but not globally L-c; we prove this for $x \geq 0$, but the same arguments work for $x \leq 0$ (the function is symmetric)
 $\delta > 0 \implies 0 \leq f(x + \delta) - f(x) = (2x + \delta)\delta \leq 3x\delta$ if $\delta \leq x$; hence, f is L-c at x with Lipschitz constant $3x$ in some (right) interval around x
 $\delta > 0 \implies f(x + \delta) - f(x) = (2x + \delta)\delta \geq 2x\delta$; hence, f cannot be L-c at x with Lipschitz constant less than $2x$, and that value is not bounded as $x \rightarrow \infty$
Symmetric arguments works for left intervals ($f(x) - f(x - \delta)$) [back]

- ▶ The standard example is $f(x) = \sqrt{|x|}$, which is easily verified to be continuous and to become “infinitely steep” as $x \rightarrow 0$, because it is the inverse function of $y = x^2$ (for $x \geq 0$), and x^2 becomes “infinitely flat” as $x \rightarrow 0$. Again, we'll have a much simpler proof later, after we present the relationships between L-c and derivatives; in fact, the proof is so much simpler that it is not worth proceeding now, we just wait until we have the right tools [back]

- ▶ Let x_* be any optimal solution in X ; by definition it belongs to (at least) one interval $[x_i, x_{i+1}]$, with $x_{i+1} - x_i \leq 2\varepsilon / L$. Assume that $x_* - x_i \leq x_{i+1} - x_*$ (the other case is analogous); then $x_* - x_i \leq \varepsilon / L$. Hence, L-c gives $f(x_i) - f(x_*) \leq L|x_i - x_*| \leq \varepsilon$ [back]

- ▶ Basically done this already: $Dr^k < \varepsilon \equiv r^k < \varepsilon / D \equiv \log(r^k) < \log(\varepsilon / D) \equiv k \log(r) < \log(\varepsilon / D) \equiv k > \log(\varepsilon / D) / \log(r)$ as $r < 1 \implies \log(r) < 0$.
Hence, $k \geq \log(D / \varepsilon) / \log(1 / r) = \log(D / \varepsilon) / (-\log(r))$
Now, $\log(1 / 0.618) \approx \log(1.618) \approx 0.21$, $1 / 0.21 \approx 4.78$ [back]

- ▶ Golden ratio search has $r = 0.5$: $1 / \log(1 / r) \approx 3.32$. But each iteration of the algorithm requires two function evaluations, so the factor is ≈ 6.64 : less iterations, but more evaluations **[back]**

- ▶ Since $f(\cdot)$ is L-c, $d^i = |x^i - x_*| \leq \delta \implies r^i = f^i - f_* \leq Ld^i \leq L\delta$. Hence, to get $r^i \leq \varepsilon$ it is sufficient to ensure that $d^i \leq \varepsilon / L$, whence the bound **[back]**

- ▶ $\lim_{t \rightarrow 0} [f(x+t) - f(x)] / t = L$ finite $\implies \lim_{t \rightarrow 0} t([f(x+t) - f(x)] / t) = \lim_{t \rightarrow 0} f(x+t) - f(x) = L \lim_{t \rightarrow 0} t = 0$ (limit of a product = product of the limits); that $\Leftarrow$ does not hold is proven by $f(x) = |x|$ **[back]**

- ▶ $f(\cdot)$ L-c $\implies |f(x+t) - f(x)| \leq L|t| \equiv |[f(x+t) - f(x)] / t| \leq L$;
 now just take the $\lim_{t \rightarrow 0}$
 Yes, the other direction is also true: by the Mean Value Theorem [6, Theorem 2.3.9], $f(z) - f(x) = f'(w)(z - x)$ for some w in the interval of extremes x and z ; take the $|\cdot|$ and use $|f'(w)| \leq L$ **[back]**

- Consider $f(x) = \sqrt[3]{x^2}$, whose derivative is $f'(x) = 2 / 3x^{1/3}$ (possibly written in the more complex but algebraic-proof form $(2x) / (3(x^2)^{2/3})$). Hence, $\lim_{x \rightarrow 0^-} f'(x) = -\infty$ and $\lim_{x \rightarrow 0^+} f'(x) = \infty$. In plain words, this is because the cubic root is the inverse function of x^3 , which is “flat in 0”; inverse functions “exchange the axes”, which means that if the graph of the function is “horizontal” as some x , then the graph of its inverse is “vertical” at the same x , which implies $f'(x) = \pm\infty$. Thus $f'(\cdot)$ is not bounded in any interval around 0, and therefore $f(\cdot)$ is not L-c there. Of course, $f'(x)$ is not continuous in 0 **[back]**
- $f'(x) = 0$ for some $x \in [\underline{x}^i, \bar{x}^i]$; $L - c$ of f' gives $|0 - f'(\underline{x}^i)| \leq L|x - \underline{x}^i|$ and $|f'(\bar{x}^i) - 0| \leq L|\bar{x}^i - x|$, whence $\min\{f'(\underline{x}^i), f'(\bar{x}^i)\} \leq L \min\{|x - \underline{x}^i|, |\bar{x}^i - x|\} \leq L\delta / 2$ (in the worst case, x is equidistant from the extremes); thus, the stopping criterion have to be satisfied when $\delta = 2\varepsilon / L$, i.e., within at most $3.32 \log(LD / 2\varepsilon)$ iterations **[back]**

- ▶ For $x_+ = \Delta x = 1$, the algorithm tries the iterates 1, 2, 4, 8, ..., i.e., 2^i . With $f(x) = \sin(\pi x + 3\pi/4) \implies f'(x) = \pi \cos(\pi x + 3\pi/4)$ we have $f'(2^i) = \pi \cos(\pi 2^i + 3\pi/4) = \pi \cos(3\pi/4) = -\pi\sqrt{2}/2 \approx -2.22$ (2^i is always even and $\cos(\cdot)$ has period 2π); that is, the algorithm always finds a “very negative” derivative and never stops, although $f(\cdot)$ has plenty of local minima. Clearly, by only very slightly changing the constants the counterexample would break down **[back]**

- ▶ $x_- f'(x_+) - x_+ f'(x_-) = x_- f'(x_+) - x_+ f'(x_-) + x_- f'(x_-) - x_- f'(x_-) = x_- (f'(x_+) - f'(x_-)) - f'(x_-)(x_+ - x_-)$. Divide by $f'(x_+) - f'(x_-)$ to get $x = x_+ + \alpha(x_+ - x_-)$ with $0 \leq \alpha = -f'(x_-) / (f'(x_+) - f'(x_-)) \leq 1$; it is then plain to see that $x_- \leq x \leq x_+$ **[back]**

- ▶ A full development would not be didactical. The four conditions are $ax_+^2 + bx_+ + c = f(x_+)$, $ax_-^2 + bx_- + c = f(x_-)$, $2ax_+ + b = f'(x_+)$, $2ax_- + b = f'(x_-)$; each three of them give a linear system with three equations in the three unknowns a, b, c that gives (not necessarily) different solutions (mind the special cases) and therefore quadratic models **[back]**

- ▶ No point to repeat [5, § 2.4.2][4, p. 57] here **[back]**
- ▶ $[Q_i]'(x) = L_i'(x) = f'(x^i) + f''(x^i)(x - x^i) = 0 \equiv$
 $x - x^i = -f'(x^i) / f''(x^i)$ **[back]**
- ▶ Since $f''(x_*) \neq 0$, $|f''(x_*)| > 0$; take e.g. $k_2 = |f''(x_*)| / 2 [> 0]$, by continuity of $f''(\cdot)$ at x_* , $\exists \delta > 0$ s.t. $|2k_2 - |f''(x)|| \leq k_2 \implies$
 $|f''(x)| \geq k_2 \forall x \in X$. Since $f'''(\cdot)$ is continuous, also $|f'''(\cdot)|$ is, hence $k_1 = \max\{|f'''(x)| : x \in X\} < \infty$ [6, Th. 2.2.9] **[back]**